JEE-2008
Paper I

1. Consider the two curves

$$C_1 : y^2 = 4x$$
$$C_2 : x^2 + y^2 - 6x + 1 = 0$$

Then,

(A) C_1 and C_2 touch each other only at one point
(B) C_1 and C_2 touch each other exactly at two points
(C) C_1 and C_2 intersect (but do not touch) at exactly two points
(D) C_1 and C_2 neither intersect nor touch each other

Answer ○ (A) ● (B) ○ (C) ○ (D)

2. If $0 < x < 1$, then

$$\sqrt{1+x^2}\ [\ \{x \cos(\cot^{-1} x) + \sin(\cot^{-1} x)\}^2 - 1\ ]^{\frac{1}{2}} =$$

(A) $\dfrac{x}{\sqrt{1+x^2}}$ (B) x (C) $x\sqrt{1+x^2}$ (D) $\sqrt{1+x^2}$

Answer ○ (A) ○ (B) ● (C) ○ (D)

3. The edges of a parallelopiped are of unit length and are parallel to non-coplanar unit vectors $\hat{a}, \hat{b}, \hat{c}$ such that

$$\hat{a} \cdot \hat{b} = \hat{b} \cdot \hat{c} = \hat{c} \cdot \hat{a} = \frac{1}{2}.$$

Then, the volume of the parallelopiped is

(A) $\dfrac{1}{\sqrt{2}}$ (B) $\dfrac{1}{2\sqrt{2}}$ (C) $\dfrac{\sqrt{3}}{2}$ (D) $\dfrac{1}{\sqrt{3}}$

Answer ● (A) ○ (B) ○ (C) ○ (D)

4. Let a and b be non-zero real numbers. Then, the equation

$$(ax^2 + by^2 + c)(x^2 - 5xy + 6y^2) = 0$$

represents

(A) four straight lines, when $c = 0$ and a, b are of the same sign
(B) two straight lines and a circle, when $a = b$, and c is of sign opposite to that of a
(C) two straight lines and a hyperbola, when a and b are of the same sign and c is of sign opposite to that of a
(D) a circle and an ellipse, when a and b are of the same sign and c is of sign opposite to that of a

Answer ○ (A) ● (B) ○ (C) ○ (D)

5. Let

$g(x) = \dfrac{(x-1)^n}{\log \cos^m (x-1)}$; $0 < x < 2$, m and n are integers, $m \neq 0$, $n > 0$, and let p be the left hand derivative of $|x-1|$ at $x = 1$.

If $\lim_{x \to 1+} g(x) = p$, then

(A) $n = 1, m = 1$
(B) $n = 1, m = -1$
(C) $n = 2, m = 2$
(D) $n > 2, m = n$

Answer ○ (A) ○ (B) ● (C) ○ (D)

6. The total number of local maxima and local minima of the function

$$f(x) = \begin{cases} (2+x)^3, & -3 < x \leq -1 \\ x^{2/3}, & -1 < x < 2 \end{cases}$$

is

(A) 0
(B) 1
(C) 2
(D) 3

Answer ○ (A) ○ (B) ● (C) ○ (D)

7. A straight line through the vertex P of a triangle PQR intersects the side QR at the point S and the circumcircle of the triangle PQR at the point T. If S is not the centre of the circumcircle, then

(A) $\dfrac{1}{PS} + \dfrac{1}{ST} < \dfrac{2}{\sqrt{QS \times SR}}$

(B) $\dfrac{1}{PS} + \dfrac{1}{ST} > \dfrac{2}{\sqrt{QS \times SR}}$

(C) $\dfrac{1}{PS} + \dfrac{1}{ST} < \dfrac{4}{QR}$

(D) $\dfrac{1}{PS} + \dfrac{1}{ST} > \dfrac{4}{QR}$

Answer ○ (A) ● (B) ○ (C) ● (D)

8. Let $P(x_1, y_1)$ and $Q(x_2, y_2)$, $y_1 < 0$, $y_2 < 0$, be the end points of the latus rectum of the ellipse $x^2 + 4y^2 = 4$. The equations of parabolas with latus rectum PQ are

(A) $x^2 + 2\sqrt{3}\ y = 3 + \sqrt{3}$

(B) $x^2 - 2\sqrt{3}\ y = 3 + \sqrt{3}$

(C) $x^2 + 2\sqrt{3}\ y = 3 - \sqrt{3}$

(D) $x^2 - 2\sqrt{3}\ y = 3 - \sqrt{3}$

Answer ○ (A) ● (B) ● (C) ○ (D)

9. Let

$$S_n = \sum_{k=1}^{n} \frac{n}{n^2 + kn + k^2} \quad \text{and} \quad T_n = \sum_{k=0}^{n-1} \frac{n}{n^2 + kn + k^2},$$

for $n = 1, 2, 3, \cdots$. Then,

(A) $S_n < \dfrac{\pi}{3\sqrt{3}}$

(B) $S_n > \dfrac{\pi}{3\sqrt{3}}$

(C) $T_n < \dfrac{\pi}{3\sqrt{3}}$

(D) $T_n > \dfrac{\pi}{3\sqrt{3}}$

Answer ● (A) ○ (B) ○ (C) ● (D)

10. Let $f(x)$ be a non-constant twice differentiable function defined on $(-\infty, \infty)$ such that $f(x) = f(1-x)$ and $f'\left(\frac{1}{4}\right) = 0$. Then,

(A) $f''(x)$ vanishes at least twice on $[0, 1]$

(B) $f'\left(\frac{1}{2}\right) = 0$

(C) $\displaystyle\int_{-1/2}^{1/2} f\left(x + \frac{1}{2}\right) \sin x\, dx = 0$

(D) $\displaystyle\int_{0}^{1/2} f(t)\, e^{\sin \pi t}\, dt = \int_{1/2}^{1} f(1-t)\, e^{\sin \pi t}\, dt$

Answer ● (A) ● (B) ● (C) ● (D)

11. Let f and g be real valued functions defined on interval $(-1, 1)$ such that $g''(x)$ is continuous, $g(0) \neq 0$, $g'(0) = 0$, $g''(0) \neq 0$, and $f(x) = g(x)\sin x$.

STATEMENT-1 : $\lim_{x \to 0} [g(x)\cot x - g(0) \text{ cosec } x] = f''(0)$.

and

STATEMENT-2 : $f'(0) = g(0)$.

(A) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 **is** a correct explanation for STATEMENT-1

(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 **is NOT** a correct explanation for STATEMENT-1

(C) STATEMENT-1 is True, STATEMENT-2 is False

(D) STATEMENT-1 is False, STATEMENT-2 is True

Answer ● (A) ○ (B) ○ (C) ○ (D)

12. Consider three planes

$P_1 : x - y + z = 1$
$P_2 : x + y - z = -1$
$P_3 : x - 3y + 3z = 2$.

Let L_1, L_2, L_3 be the lines of intersection of the planes P_2 and P_3, P_3 and P_1, and P_1 and P_2, respectively.

STATEMENT-1 : At least two of the lines L_1, L_2 and L_3 are non-parallel.

and

STATEMENT-2 : The three planes do not have a common point.

(A) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 **is** a correct explanation for STATEMENT-1

(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 **is NOT** a correct explanation for STATEMENT-1

(C) STATEMENT-1 is True, STATEMENT-2 is False

(D) STATEMENT-1 is False, STATEMENT-2 is True

Answer ○ (A) ○ (B) ○ (C) ● (D)

13. Consider the system of equations

$$x - 2y + 3z = -1$$
$$-x + y - 2z = k$$
$$x - 3y + 4z = 1 .$$

STATEMENT-1 : The system of equations has no solution for $k \neq 3$.

and

STATEMENT-2 : The determinant $\begin{vmatrix} 1 & 3 & -1 \\ -1 & -2 & k \\ 1 & 4 & 1 \end{vmatrix} \neq 0$, for $k \neq 3$.

(A) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 **is** a correct explanation for STATEMENT-1

(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 **is NOT** a correct explanation for STATEMENT-1

(C) STATEMENT-1 is True, STATEMENT-2 is False

(D) STATEMENT-1 is False, STATEMENT-2 is True

Answer ● (A) ○ (B) ○ (C) ○ (D)

14. Consider the system of equations
$ax + by = 0$, $cx + dy = 0$, where $a, b, c, d \in \{0, 1\}$.

STATEMENT-1 : The probability that the system of equations has a unique solution is $\frac{3}{8}$.

and

STATEMENT-2 : The probability that the system of equations has a solution is 1 .

(A) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 **is** a correct explanation for STATEMENT-1

(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 **is NOT** a correct explanation for STATEMENT-1

(C) STATEMENT-1 is True, STATEMENT-2 is False

(D) STATEMENT-1 is False, STATEMENT-2 is True

Answer ○ (A) ● (B) ○ (C) ○ (D)

Paragraph for Question Nos. 15 to 17

A circle C of radius 1 is inscribed in an equilateral triangle PQR. The points of contact of C with the sides PQ, QR, RP are D, E, F, respectively. The line PQ is given by the equation $\sqrt{3}\,x + y - 6 = 0$ and the point D is $\left(\frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$. Further, it is given that the origin and the centre of C are on the same side of the line PQ.

15. The equation of circle C is

(A) $(x-2\sqrt{3})^2+(y-1)^2=1$ (B) $(x-2\sqrt{3})^2+(y+\frac{1}{2})^2=1$

(C) $(x-\sqrt{3})^2+(y+1)^2=1$ (D) $(x-\sqrt{3})^2+(y-1)^2=1$

Answer ○ (A) ○ (B) ○ (C) ● (D)

16. Points E and F are given by

(A) $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}\right), (\sqrt{3}, 0)$ (B) $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right), (\sqrt{3}, 0)$

(C) $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}\right), \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ (D) $\left(\frac{3}{2}, \frac{\sqrt{3}}{2}\right), \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

Answer ● (A) ○ (B) ○ (C) ○ (D)

17. Equations of the sides QR, RP are

(A) $y=\frac{2}{\sqrt{3}}x+1,\ y=-\frac{2}{\sqrt{3}}x-1$ (B) $y=\frac{1}{\sqrt{3}}x,\ y=0$

(C) $y=\frac{\sqrt{3}}{2}x+1,\ y=-\frac{\sqrt{3}}{2}x-1$ (D) $y=\sqrt{3}\,x,\ y=0$

Answer ○ (A) ○ (B) ○ (C) ● (D)

Paragraph for Question Nos. 18 to 20

Consider the functions defined implicitly by the equation $y^3-3y+x=0$ on various intervals in the real line.

If $x \in (-\infty, -2) \cup (2, \infty)$, the equation implicitly defines a unique real valued differentiable function $y = f(x)$.

If $x \in (-2, 2)$, the equation implicitly defines a unique real valued differentiable function $y = g(x)$ satisfying $g(0)=0$.

18. If $f(-10\sqrt{2}) = 2\sqrt{2}$, then $f''(-10\sqrt{2}) =$

(A) $\frac{4\sqrt{2}}{7^3 3^2}$ (B) $-\frac{4\sqrt{2}}{7^3 3^2}$ (C) $\frac{4\sqrt{2}}{7^3 3}$ (D) $-\frac{4\sqrt{2}}{7^3 3}$

Answer ○ (A) ● (B) ○ (C) ○ (D)

19. The area of the region bounded by the curve $y = f(x)$, the x-axis, and the lines $x = a$ and $x = b$, where $-\infty < a < b < -2$, is

(A) $\int_a^b \frac{x}{3\left((f(x))^2 - 1\right)} dx + bf(b) - af(a)$

(B) $-\int_a^b \frac{x}{3\left((f(x))^2 - 1\right)} dx + bf(b) - af(a)$

(C) $\int_a^b \frac{x}{3\left((f(x))^2 - 1\right)} dx - bf(b) + af(a)$

(D) $-\int_a^b \frac{x}{3\left((f(x))^2 - 1\right)} dx - bf(b) + af(a)$

Answer ● (A) ○ (B) ○ (C) ○ (D)

20. $\int_{-1}^{1} g'(x)\, dx =$

(A) $2g(-1)$ (B) 0 (C) $-2g(1)$ (D) $2g(1)$

Answer ○ (A) ○ (B) ○ (C) ● (D)

Paragraph for Question Nos. 21 to 23

Let A, B, C be three sets of complex numbers as defined below

$$A = \{z : \operatorname{Im} z \geq 1\}$$
$$B = \{z : |z - 2 - i| = 3\}$$
$$C = \{z : \operatorname{Re}((1 - i)z) = \sqrt{2}\}.$$

21. The number of elements in the set $A \cap B \cap C$ is

(A) 0 (B) 1 (C) 2 (D) ∞

Answer ○ (A) ● (B) ○ (C) ○ (D)

22. Let z be any point in $A \cap B \cap C$. Then, $|z + 1 - i|^2 + |z - 5 - i|^2$ lies between

(A) 25 and 29 (B) 30 and 34 (C) 35 and 39 (D) 40 and 44

Answer ○ (A) ○ (B) ● (C) ○ (D)

23. Let z be any point in $A \cap B \cap C$ and let w be any point satisfying $|w-2-i|<3$ Then, $|z|-|w|+3$ lies between

(A) -6 and 3 (B) -3 and 6 (C) -6 and 6 (D) -3 and 9

Answer

○ ● ○ ○
(A) (B) (C) (D)

OR

○ ○ ● ○
(A) (B) (C) (D)

OR

○ ○ ○ ●
(A) (B) (C) (D)

24. Students I, II and III perform an experiment for measuring the acceleration due to gravity (g) using a simple pendulum. They use different lengths of the pendulum and/or record time for different number of oscillations. The observations are shown in the table.

Least count for length = 0.1 cm

Least count for time = 0.1 s

Student	Length of the pendulum (cm)	Number of oscillations (n)	Total time for (n) oscillations (s)	Time period (s)
I	64.0	8	128.0	16.0
II	64.0	4	64.0	16.0
III	20.0	4	36.0	9.0

If E_I, E_{II} and E_{III} are the percentage errors in g, i.e., $\left(\frac{\Delta g}{g} \times 100\right)$ for students I, II and III, respectively,

(A) $E_I = 0$

(B) E_I is minimum

(C) $E_I = E_{II}$

(D) E_{II} is maximum

Answer

○ ● ○ ○
(A) (B) (C) (D)